

Tradelatest Book — Grok Review Delta

Chapters updated for the review: Quick Start, remediations, INOUT, Program 9, A2

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Grok review pass export — reading convenience only.
On conflict, repository code/config wins.

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Chapter 00 - Quick Start: One Candle's Journey

Part I - Foundations

Status of this chapter: Written (Grok review pass, 2026-08-07)

Why this chapter exists

The rest of this book is deliberately careful about *how* knowledge is classified, which record system owns what, and what remains unfixed. That meta-layer is load-bearing for contributors who will edit the repo - but it can overwhelm a reader who only needs to understand **what the trading system does**. This chapter short-circuits the meta-layers and gives the core mechanics in one pass.

What problem it solves

"I just need to understand the spine" without reading Chapters 3, 17, and 18 first.

What you need to already know

Nothing. Jump back to [Chapter 1](#) when you want the *why*, or [Chapter 2](#) for the full invariant set.

The idea

The system in one sentence

Tradelatest takes **M15 OHLCV candles**, scores each bar through **four independent engines** (CRT, Gaussian, Zone Gate, RR), **fuses** those scores, **decides** accept/reject on semantic grounds, plans **entry geometry**, and finally allows or blocks the order through a **capital risk gate**. Everything is file-backed: no database, no message broker, no cloud dependency.

The happy path (one candle)

Step	What happens	Where it lives
1	Load candle (CSV / MT5 / live feed)	<code>src/inout/*_candle_fetcher.py</code> , loaders in runtime
2	Build a 39-dim feature vector (no lookahead)	<code>src/features/feature_pipeline.py</code>
3	CRT state machine advances (RANGE->...->EXECUTION)	<code>src/config_layer/crt_engine_v2.py</code>
4	Four engines score the bar independently	<code>src/engines/*</code> via <code>EngineRunner</code>
5	Fusion combines scores under a completeness gate	<code>src/core/fusion_engine.py</code>
6	Decision Engine: semantic approval only	<code>src/core/decision_engine.py</code>
7	Execution Planner: intent + entry; CRT owns SL/TP levels	<code>execution_planner.py</code> + <code>compute crt levels</code>

8	Ultron Risk Gate: final capital / RR / exposure check	src/core/ultron_risk_gate.py
9	Live path: HookedLiveEngine.process on each tick	src/runtime/live_engine_hook.py

Backtests stream candle-by-candle through the same spine (src/runtime/backtest_v2.py). Live always runs the fusion gate; research/backtest gate mode is config-driven (backtest.engine_gate_enabled).

Priority order (not "profit first")

The system's declared priorities are, in order: **replay correctness -> explainability -> telemetry -> AI as advisory only -> profit**. Profit is a consequence of a trustworthy process, not a license to skip gates. See [Chapter 1](#) for the full argument.

What is actually load-bearing today

Surface	Status (honest)
CRT state machine (mechanism)	CLOSED for OHLCV -> TRADE_OPENED boundary
Feature code surface (39-dim)	CLOSED for formula/PIT/write-site hygiene - not economic proof
Gaussian engine	AUDITED - live score collapses near-constant ~0.8825 (F-060)
Zone Gate	AUDITED - geometric HARD gate; non-pivotal on fusion decisions (F-036)
RR / rr_fusion	AUDITED - base RR flows; rr_fusion disabled (F-038/F-044)
BitNet / TradeNet	Built; inert or unwired on the active production config
Research Programs 1-8	CLOSED nulls under a fixed expectancy bar - discipline working
Active production config	configs/production/ACTIVE_VERSION -> currently v2_multi_2026_04 on this branch

How config becomes production

1. Tunables live in configs/production/*.json - no magic numbers in hot-path Python.
2. ConfigValidator.validate() runs hard + soft quality gates -> ValidationReport.
3. Only decision == "APPROVE" may promote; SHA-256 hash + append-only promotion_log.jsonl.
4. Rollback = restore archived config + update ACTIVE_VERSION.

Details: [Chapter 16](#).

How to read the rest of the book

If you want...	Read
Full why + invariants	Ch.01 -> Ch.02
Ontology, features, CRT depth	Ch.06 -> Ch.08

Engines, fusion, decision honesty	Ch.10 -> Ch.12
Geometry + risk + live path	Ch.13 -> Ch.15
Governance / findings / closure	Ch.16 -> Ch.18
Research programs + numeric results table	Ch.19 -> Ch.20 (Part VII; also dedicated PDF *Research-Grok.pdf)
In-repo agent + multi-LLM ops	Ch.21 -> Ch.23
Open defects & book gaps	A2

Rule on conflict: this book is a map. Code and active config win.

Known defects you should not ignore (even on a quick read)

1. **ATR unit mismatch** in `compute_crt_levels` - SL buffer may treat relative ATR as price units ([Ch.13](#) has proposed remediations; unfixed pending owner decision).
2. **CRTConfig split-brain (F-057)** - programmatic `BacktestRunner(cfg)` without `crt_config` can miss production JSON overrides.
3. **Doc-drift elsewhere in the repo** - e.g. some reference docs still say 38-dim; live code is 39-dim. This book tracks code.

Classification

Concept	Status
Candle -> order spine	Production
This Quick Start chapter	Orientation aid - not an authority surface

Authoritative sources

- `docs/architecture/signal-flow.md` - the 7-step walk with failure modes.
- `docs/architecture/goal.md` - purpose, invariants, priority order.
- `configs/production/ACTIVE_VERSION` - Tier-0 runtime truth.
- `src/core/engine_runner.py` - orchestrator every decision path passes through.

Unresolved questions

None for this chapter - open items live in [A2](#).

Previous: [Table of Contents](#) * **Next:** [Chapter 01 - Why Tradelatest Exists](#)

Related: [Chapter 02 - The Invariants and the Happy Flow](#)

Memory: `docs/memory/architecture-memory.md` before editing the spine.

Chapter 13 - The Execution Planner: From Decision to Order Geometry

Part V - Execution

Status of this chapter: Written

Why this chapter exists

An ACCEPT from the Decision Engine (Chapter 12) is still just an approval - it's not yet an order. Something has to turn "yes, trade this" into a concrete entry price, and classify *what kind* of trade this is. This chapter covers that step, and flags two live, unresolved issues that anyone touching this code should know about before assuming the geometry it produces is trustworthy.

What problem it solves

Explains what `ExecutionPlannerV1_2` actually computes - and, just as importantly, what it deliberately does *not* compute.

What you need to already know

[Chapter 12](#) - the planner only runs on an ACCEPT decision.

The idea

A classifier, not a geometry engine

`src/config_layer/execution_planner.py`'s `ExecutionPlannerV1_2.plan()` is, at its core, an intent classifier plus a gate. It sorts an accepted setup into one of four intents - BREAKOUT, PULLBACK, LIQ_SWEEP, REVERSAL - derives an entry price appropriate to that intent, and delegates final approval to `GateIntelligence`. **Stop-loss, take-profit, and reward:risk are explicitly not computed here.** That geometry comes from a different module entirely: CRT's own `compute crt levels` (inside `crt_engine_v2.py`), which owns the SL/TP math because it's the module that understands the structural levels (sweep point, displacement extent) the stop and target are actually measured against.

An open architectural caveat: which config the planner's geometry actually sees

There is a documented split-brain in how `CRTConfig` gets resolved on the *programmatic* path (tests, the tuner, embedders - anywhere `BacktestRunner(cfg)` is called with `crt_config=None`): in that case, `ConfigBuilder.build()` runs with no overrides, and the router's hardcoded FOREX/CRYPTO default profiles win - the production JSON config is never actually opened. Concretely, `expansion_atr_min_distance` ends up at 0.08 on this path versus 0.30 in the production JSON - a 3.75x difference on the very gate that governs the DISPLACEMENT->EXPANSION transition ([Chapter 8](#)'s state machine). The CLI entry point is unaffected - this is specifically a programmatic-construction gap. It's open, not yet fixed, and worth knowing about before trusting a programmatically-run backtest's structural-gate behavior at face value.

A second open caveat: SL geometry may be using the wrong units

A separate, still-unresolved issue concerns `compute crt levels` itself: it appears to treat the `atr` feature as if it were already in price units when computing the stop-loss buffer, but - per [Chapter 7](#)'s feature pipeline - `atr` in the canonical vector is close-relative, not an absolute price quantity. On XAUUSD this produces stop-loss buffers on the order of thousands of times too small (sub-cent stops on LIQ_SWEEP entries), and the same unit

confusion is implicated in unbounded position sizing downstream. This is a real, identified, **currently unfixed** issue awaiting an explicit user decision on remediation - it is not a hypothetical, and it is not something this book fixes as a side effect of documenting it.

Proposed remediation (Grok review pass - options only; no code in this book)

The review of this book correctly noted that flagging without a path feels passive. Below are **proposal options with trade-offs**. None of these is authorized by this chapter: authority still requires an explicit owner decision + construction protocol + parity proof ([Chapter 16](#) Authority Ladder).

ATR unit mismatch - proposed options

Option	What it does	Pros	Cons / risks
A. Multiply by close	Treat pipeline <code>atr</code> as relative; convert at use-site: <code>atr_price = atr_rel * close</code> inside <code>compute_crt_levels</code> (and any sibling buffer sites)	Surgical; matches the known closed form used elsewhere for FM-022/023-style defects; easy to parity-test	Must inventory every consumer of the same <code>atr</code> key in geometry; risk of double-scaling if any caller already converts
B. Pass absolute ATR separately	Geometry API takes <code>atr_abs</code> (from true-range SMA in price units) and never reads the relative feature for stops	Clean separation of feature vector vs geometry units; hard to misuse	New argument surface through CRT geometry callers; dual sources of ATR can drift if periods differ
C. Config-gated dual path	<code>geometry_atr_basis:</code> <code>relative_legacy \ </code> <code>absolute_price</code> with default = current behavior (byte-identical)	Matches F-061/F-066 pattern (strict require + default freeze); enables A/B shadow	Longer-lived dual code; still needs freeze-pin re-certification before activating on production

Recommended default for a future implement turn: C then A - land the conversion behind a strict config key with `relative_legacy` default (hash-neutral), prove ledger parity, then open a shadow config that flips to absolute and measures $\Delta G001$ before any promotion. Do **not** flip the active production config in the same turn as the wiring.

F-057 CRTConfig programmatic split-brain - proposed options

Option	What it does	Pros	Cons / risks
1. Always load ACTIVE_VERSION	When <code>crt_config</code> is <code>None</code> , <code>BacktestRunner</code> / <code>ConfigBuilder</code> loads production JSON via <code>get_prod_config()</code> instead of bare FOREX/CRYPTO profiles	Closes the 3.75x <code>expansion_atr_min_distance</code> gap; CLI and programmatic paths agree	Any test that silently relied on hardcoded profiles will change behavior - needs a test inventory
2. Fail-closed	Raise if <code>crt_config</code> is <code>None</code> on programmatic construction	Forces every caller to be explicit	Breaks tuner/tests until each injects config; noisier migration
3. Explicit profile enum	Require <code>profile="production" \ "router_default"</code> - no implicit default	Makes the choice auditable	API churn; still easy to pick the wrong profile

Recommended default for a future implement turn: 1 with a loud WARN log on the migration window, plus a focused regression that asserts programmatic build equals CLI-loaded production for the structural gates that F-057 names. Treat as backtest Phase-2 work (as the finding already scopes it) - not a silent one-line patch.

What this book still will not do

These remediations are **not implemented by documenting them**. Implementing either issue is a governed code change (construction protocol, tests, parity, owner approval on behavior change). Status remains: **OPEN / awaiting decision**.

Classification

Concept	Status
ExecutionPlannerV1_2.plan() intent classification	Production
SL/TP geometry (compute_crt_levels, lives in CRT engine)	Production, known defect, unfixed (ATR unit mismatch)
Programmatic CRTConfig resolution (non-CLI paths)	Production, known split-brain, OPEN (F-057)

Authoritative sources

- `src/config_layer/execution_planner.py` - ExecutionPlannerV1_2.plan().
- `src/config_layer/crt_engine_v2.py` - compute_crt_levels (the SL/TP/RR geometry, owned by CRT not the planner).
- `docs/topics/execution-planning.md` - the always-synced topic doc.
- `docs/current-findings.md` F-057 - the CRTConfig programmatic split-brain, in full.
- `docs/topics/ai-automation-agent.md` §GateIntelligence cross-references (if present) - the approval delegate.

Unresolved questions

- **The ATR unit mismatch in compute_crt_levels** - identified, scoped, proposed above (options A/B/C); still awaiting owner pick + implement authority. Rolled up in [A2](#).
- **F-057's CRTConfig split-brain** - status OPEN; proposed options 1/2/3 above; still "backtest Phase 2" not implemented as of this writing.

Previous: [Chapter 12 - The Decision Engine](#) * **Next:** [Chapter 14 - Ultron Risk Gate](#)
Related: [Chapter 08 - The CRT State Machine](#) (owns the SL/TP math this chapter describes)
Memory: `docs/memory/architecture-memory.md`.

Chapter 15 - Live Execution and INOUT

Part V - Execution

Status of this chapter: Written (INOUT ambiguity **resolved** in Grok review pass, 2026-08-07)

Why this chapter exists

Everything in Parts II-V so far describes the backtest path faithfully, and the live path *mostly* the same way - but "mostly" is doing real work in that sentence, and this chapter is where the book is most honest about the gap between the two. It closes Part V by describing what actually runs on a live tick, and by resolving a naming collision that earlier drafts left open: `src/inout/` vs `archive/inout_legacy/`.

What problem it solves

Distinguishes "what the backtest replays" from "what a live tick actually triggers," and is explicit about which of the two the rest of this book's Parts II-V chapters were describing (mostly the former, since it's the better-evidenced, more heavily tested path).

What you need to already know

[Chapter 14](#) - the one point where a live-execution rail rejoins the synchronous spine.

The idea

The live tick entry point

The confirmed entry point for a live candle is `HookedLiveEngine.process` in `src/runtime/live_engine_hook.py:582`. This is what actually gets called on a live tick, and it draws on `src/live/`, `src/inout/`, and `src/execution/loop.py` as supporting modules.

`src/execution/loop.py` - real code, no caller yet

`src/execution/loop.py` (plus `alert_manager.py`, `override_handler.py`) implements a multi-signal execution loop, but per the repository's own topic tracking it is explicitly **scaffolding - no caller yet**. It has tests (`test_execution_loop`) but isn't wired into a live path that actually invokes it.

INOUT naming collision - RESOLVED (Grok review pass, 2026-08-07)

Two paths share the word "inout" but are **not the same subsystem**. Direct comparison of both trees resolves the earlier open question without inventing a third story:

Path	What it actually is	Status
<code>archive/inout_legacy/ARCHIVED_2026-05-02</code>	Parallel execution pipeline that bypassed the canonical spine: <code>scanner.py</code> , <code>state_machine.py</code> , <code>controller.py</code> , <code>executor.py</code> , <code>runner.py</code> , <code>probability_engine.py</code> , plus a <code>SQLite db.py</code>	Legacy / Archived 2026-05-02 - README states it violated spine + no-DB conventions; no production <code>src/</code> importers at archival time

src/inout/	Market-data I/O only - candle/funding fetchers: <code>mt5_candle_fetcher.py</code> , <code>alphavantage_candle_fetcher.py</code> , <code>hummingbot_candle_fetcher.py</code> , <code>perp_funding_fetcher.py</code> (4 modules + package cache)	Production data-ingestion helpers - not a strategy rail, not a second decision path
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Verdict: `src/inout/` is **not** the successor of the archived INOUT rail. It is a **name reuse** for a different responsibility (fetch OHLCV / funding into CSV loaders). The archived rail's replacement for *execution* is stated in its own README: all valid live orchestration routes through `src/runtime/live_engine_hook.py -> HookedLiveEngine.process()`. CLAUDE.md's placement note that "live-mode code" may land under `src/inout/` is therefore best read as historical folder convention for I/O, not an invitation to revive the archived parallel spine.

`docs/architecture/goal.md`'s phrase "INOUT currently archived" correctly describes the **strategy rail** now under `archive/inout_legacy/`. It does **not** mean the current `src/inout/` fetchers are dead.

Classification

Concept	Status
<code>HookedLiveEngine.process</code> (live tick entry point)	Production
<code>src/execution/loop.py</code> (multi-signal execution loop)	Partial - built, tested, no live caller
INOUT strategy rail (<code>archive/inout_legacy/</code>)	Legacy / Archived (2026-05-02)
<code>src/inout/</code> data fetchers	Production (I/O) - distinct from the archived rail

Authoritative sources

- `src/runtime/live_engine_hook.py:582` - `HookedLiveEngine.process`.
- `src/execution/loop.py`, `alert_manager.py`, `override_handler.py`.
- `docs/topics/live-execution.md`, `docs/topics/execution-loop.md` - the always-synced topic docs.
- `docs/architecture/goal.md` §2 - "INOUT currently archived" (strategy rail, not fetchers).
- `archive/inout_legacy/ARCHIVED_2026_05_02/README.md` - archival reason + replacement path.
- `src/inout/*.py` - current fetcher modules (MT5 / Alpha Vantage / Hummingbot / perp funding).

Unresolved questions

- Whether residual `tests/inout/` still exercise the archived pipeline (the archive README suggested moving them to `tests/archive/`) was not re-audited this pass - hygiene only, not a live-path risk.
- `src/execution/loop.py` remains scaffolding with no live caller (unchanged).

Previous: [Chapter 14 - Ultron Risk Gate](#) * **Next:** [Chapter 16 - Config-First Doctrine and the Promotion Path](#)
Related: [Chapter 02 - The Invariants and the Happy Flow](#) (where INOUT was first introduced as a kitchen feeder)
Memory: `docs/memory/runtime-memory.md`.

Chapter 19 - The Research Programs: Falsification as a Discipline

Part VII - Research

Status of this chapter: Written (research data table expanded Grok pass)

Why this chapter exists

Chapter 1 promised that pattern interpreters and, by extension, trading ideas in general are "measured, never asserted." This chapter is where that promise gets tested at scale: a numbered sequence of Research Programs, each pre-registered, each run to a decisive conclusion, and - almost without exception so far - each one closing as an honest null result. That's not a failure of the research effort. It's the effort working exactly as designed.

What problem it solves

Gives a reader the shape of the research history **with concrete result data** (not only slogans): what was tested, sample sizes / key metrics where registered, the verdict, and why "0 PROMOTE" across program after program is a *credible* finding rather than a broken research pipeline.

What you need to already know

Chapter 16's Authority Ladder - every program below is that ladder applied to a specific hypothesis: information existing is not the same as economic value, which is not the same as earned authority.

The idea

Research results at a glance (data table)

Primary authority for every row: docs/current-findings.md (findings F-0xx below). Shared measurement standard for economic arms unless noted: **intraday fixed exit model + 12 bps** round-trip cost, absolute expectancy gate (E must clear 0 OOS to PROMOTE), plus controls / permutations as each program pre-registered. All economic cells below are **research-only** - none of these results grants production fusion weight.

Prog	Question (one line)	Key data (registered)	Verdict	Finding
1	Any next-bar directional edge on crypto majors?	Toys: baseline_delta vs random ~0 (-0.076...+0.033); spine 5-13 trades/inst -> n<30; pooled n=30 E=-0.399	0 PROMOTE / KILLED	F-019
1b	Conditional pockets (sessionxvolxmom, h<=20)?	25/25 partitions BH-significant but IG~0.001-0.0036 bits; Stage-3 economics on 74 cells -> 0 pockets	ENTROPY_LEAD_NO ECON	F-020

1c	Spine RETEST selection skill beyond session?	Selected–rejected $\Delta E = +1.17R$ but SESSION=ALL rejects; ZONE 0/110; SCORE $p = 0.327$	Session filter only	F-021
1d	Exit/cost grid rescue expectancy?	Universe $n = 53,629$; every SLxTP cell $E_{oos} < 0$; best -0.271 ; reality_gap $+4.16R$ (entry info, not exit)	Exit is not the lever	F-025
2	Completed sweep->disp->retest > bare sweep?	Funnel retest $n = 47$ (~1% completion); excursion_asym(h8) = -0.319 ; loses all 4 controls	INSUFFICIENT_POWER / FROZEN	F-026
3	H1/H4 rescue directional edge?	0 PROMOTE H1 and H4; H1 E_{oos} $-0.136 \dots -0.266$; H4 expansion_breakout ~gross 0 then pooled -0.0009	CLOSED null	F-027
P&F	Point-and-figure double-top/bottom edge?	BNB $n = 8554$, PF = 0.528, $E = -0.454$, WR = 0.327; loses random_uniform $\Delta E = -0.039$	REJECT / FROZEN	F-028
4	Vol-regime LEVEL consumable directionally?	$H_{atr} = 0.885$ (vol memory real); 0 REGIME_EXPLOITABLE ; toys $n = 3.7k-35k$ still $E \sim -0.14 \dots -0.35$	Informative != money	F-030
4b	Markov P^H transition forecast?	0 REGIME_EXPLOITABLE / 15 scopes; toys REGIME_REDUNDANT vs level+lag	CLOSED redundant	F-043
4d	MTF compression->expansion (Stage1 info / Stage2 econ)?	Stage1 UNIVERSAL $p \sim 0.0005$; Stage2 directional consumer 0 PROMOTE (crypto $E -0.25 \dots -0.60R$)	Info yes / money no	F-040
5	Cross-sectional relative-value monetizable?	5 interpreters, $n = 722-8,758$; momentum PF = 0.335; 0 PROMOTE	KILLED (panel axis)	F-032
6	Carry/basis as signal on spot dispersion?	8 interpreters $n = 722-4,373$; $E \in [-0.0032, -0.0016]$; 0 PROMOTE	KILLED	F-033

6b	Funding cashflow harvest net of costs?	4 harvest_full REJECT n=104-729; pre-cost +1...+6 bps < ~24 bps RT cost	KILLED / STOP	F-034
FX	Program-1 null generalizes off crypto?	5 FX majors; toys n=7.7k-17k PF 0.02-0.10 E -1.5...-2.8R; 0 PROMOTE	Null generalizes	F-035
8	Weekly Mon-Fri liquidity-sweep ontology (FX)?	6 scopes; pooled n=512 E=-0.634R PF=0.457 ; 0 PROMOTE	CLOSED null	F-042
7	Open Interest	Named out-of-scope for P6; not run in findings ledger as of book sources	NOT RUN	-
9	M5 multi-TF expansion + OCO straddle	Pre-reg frozen; on-disk results/research/m5_mtf/ has resample_parity PASS only; Stage-1/2 verdicts not sealed into findings	PRE-REGISTERED	H-016 / prereg

How to read the table: "0 PROMOTE" means *no cell cleared the fixed absolute-expectancy bar under the pre-registered cost/exit model* - not "the research code failed." Several rows are statistically informative (entropy, vol memory, Stage-1 transition IG) yet economically dead for spot directional (or carry) constructions. That split is the point of the Authority Ladder.

Where the raw artifacts live (examples; full paths in each finding's Evidence field):

Family	Typical artifact roots
Program 1 qualify	results/research/qualification/
Conditional entropy	results/research/phase_b/
Selection effect	results/research/phase_s/
Exit grid	results/research/phase_d/
Structural asymmetry	results/research/phase_e/
HTF qualify	results/research/qualification_htf/
Regime / transition	results/research/regime/, results/research/candle_state/
Cross-section / carry	results/research/cross_sectional_*.json, carry_*.json, harvest_*.json
Weekly sweep	results/research/weekly_sweep/
Program 9 M5	results/research/m5_mtf/

The gate every program runs through

Nearly every program in this ledger is tested against the same standard: a pre-registered hypothesis, an intrabar-realistic exit model with real transaction costs (not close-only fills), and a decision rule that requires *absolute* expectancy above zero out-of-sample - not just beating a control. A program is only allowed to "PROMOTE" a finding into production authority if it clears that bar; anything else is REJECT, INSUFFICIENT_POWER, or REDUNDANT (statistically real but fully explained by something already known).

The ladder, program by program (narrative)

- **Program 1 - Next-Bar Directional Ontology.** The foundational sweep: is there *any* directional edge in next-bar entries across crypto majors, at any conditioning level (session, volatility, momentum)? Ran through qualification, conditional-entropy pocket search, a selection-effect isolation, a full trade-anatomy audit, and an exit/cost grid - every stage closed null. **CLOSED / KILLED.** This is the foundation the rest of the ladder builds on.
- **Program 2 - Structural Asymmetry.** Does a *completed* sweep->displacement->retest sequence add forward asymmetry beyond a bare sweep? One experiment, a negative point estimate, badly underpowered (n~47, roughly 1% funnel completion). **FROZEN after one experiment** - explicitly not run further until power improves.
- **Program 3 - Higher-Timeframe Directional Ontology.** Does moving to H1/H4 rescue the directional edge Program 1 killed at M15? No - zero promotions at either timeframe. **CLOSED.**
- **Program 4 / 4b - Volatility Regime and Transition.** Does contemporaneous volatility-regime *level*, or a forward Markov *transition* forecast, offer a consumable directional edge? Both informative (volatility genuinely has memory) but neither consumable - the transition-forecast variant (4b) turned out to be statistically redundant with plain current-volatility level, adding nothing beyond what was already known. **CLOSED**, both variants.
- **Program 5 - Cross-Sectional Dispersion.** The first program to leave the per-instrument frame entirely and ask whether relative-value/market-neutral positioning across instruments is monetizable. All five interpreters tested REJECT, well-powered. **KILLED.**
- **Program 6 / 6b - Carry / Basis and Carry Harvest.** Is funding-rate carry a usable signal (6), or - separately - is the raw cash-and-carry payoff itself harvestable net of costs (6b)? Both null: all eight carry/basis interpreters REJECT, and all four harvest constructions fail to clear their own turnover cost even before considering price drag. **Both KILLED**, with an explicit stop called after 6b rather than continuing to iterate on cost assumptions.
- **Program 7 - Open Interest.** Named as explicitly out of scope for Program 6 (a distinct payoff), but not yet run as of the evidence this chapter draws on.
- **Program 8 - Weekly Liquidity-Sweep Ontology.** A genuinely new calendar-locked geometry (weekly accumulation/sweep/reversal, distinct from Program 1/2's local-structural frame) tested on FX majors. Zero promotions across all six scopes tested. **CLOSED.**
- **Program 9 - M5-base Multi-TF Expansion Forecasting (Non-Directional Ontology).** See dedicated section below (Grok review pass). **Pre-registered; research authority only.**
- **A cross-asset generalization check (not itself numbered as a Program):** does Program 1's crypto-majors null generalize to FX majors? Yes - the same entry-information null replicated on five FX majors with well-powered, decisively negative toy results.

Program 9 in depth (Grok review pass - closes the coverage gap)

Earlier book drafts only noted Program 9 from commit history. The frozen pre-registration is authoritative: docs/research/preregistration-program-9.md (`../research/preregistration-program-9.md`).

Why it exists. Program 4 (F-040) found expansion forecasting **informative but not consumable** by a spot *directional* consumer. Its reopen clause allows only new data, a new domain, or a new

non-directional ontology - never parameter archaeology. Program 9 exercises two reopen keys:

- 1. **New data:** strict M5 corpus (Binance crypto ~2y; MT5 FX ~270d) that did not exist when F-040 registered ("no M5 on disk").
- 2. **New ontology:** a both-sided stop-entry **OCO straddle** at compression-box edges (long-volatility payoff), which spot long/short cannot express.

Two-stage gate (same Authority Ladder as every other program):

Stage	Question	Authority level
Stage 1	Does the M5^M15^H1^H4 conjunction carry <i>incremental</i> information beyond the M15-base key at the same instant? (Gates A raw IG + B within-K15-cell incremental)	Level 1 - information only
Stage 2	Sole consumer <code>compression_box_straddle</code> through the unchanged M4 QualificationGate	Level 2 - economic

Stage 2 runs **only if** Stage 1 clears. **Gate A pass + Gate B fail** ⇒ **M15_REDUNDANT = Stage-1 FAIL** (F-043's "real but already-known" lesson, renamed for this program).

Frozen construction highlights (D1-D7): reject-bar both-edges cancel (no invented touch order); no fill-bar TP credit; unfilled straddles excluded from n; first-compression-bar gating only; chronological 50/50 stability split; sole-consumer rule - no straddle-v2 archaeology inside Program 9.

Closure rule: Program 9 is **CLOSED** once hypotheses 9a (vol expansion), 9b (range expansion), and 9c (compression->expansion transition) each resolve to Stage-1 FAIL (incl. **M15_REDUNDANT**) **or** Stage-1 PASS + Stage-2 sole-consumer verdict - under the fixed thresholds. No within-program iteration.

Status at book time: pre-registration is frozen and present in-repo; hypothesis registry seed tracks H-016 as open/pre-reg. This book does **not** invent Stage-1/Stage-2 numeric results - verify artifacts under `results/research/m5_mtf/` (if present) and `docs/current-findings.md` before claiming a PROMOTE/REJECT. Authority remains **research/docs only** until a sealed finding says otherwise.

Related implementation surfaces (orientation, not a full walk): `src/research/candle_state/` (M5 Stage-1 kernels), `src/research/hypotheses/compression_box_straddle.py` (Stage-2 consumer), `src/research/resample.py` (calendar OHLCV resampler shared with Program 3).

Why this is the right outcome, not a broken pipeline

Every one of these closures follows the same discipline as [Chapter 9's](#) Interpreter Contract: pre-register the test, apply a fixed, honest cost/exit model, require *absolute* expectancy, and accept the answer even when it's negative. A research program that only ever produced PROMOTE verdicts on this codebase's own history would be the actual red flag - it would mean the bar wasn't being held. Programs 1 through 8 closing null, one after another, under the same rigorous standard, is what a functioning falsification discipline looks like from the outside.

Classification

Concept	Status
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Programs 1, 2, 3, 4, 4b, 5, 6, 6b, 8	Research - all CLOSED/KILLED/FROZEN , null or insufficient-power results
Program 9 (M5 multi-TF non-directional straddle)	Research - PRE-REGISTERED ; Stage results not asserted here
Program 7 (Open Interest)	Named / out-of-scope for Program 6; not independently verified as run
FX cross-asset generalization of Program 1's null	Research - confirmed generalizes

Authoritative sources

- `docs/current-findings.md` - the Funding Ledger section, Programs 1 through 8, each with its own F-0xx findings and confidence levels - this chapter's primary source, summarized not reproduced.
- `docs/research/preregistration-program-9.md` - Program 9 frozen pre-registration (primary for P9).
- `docs/research-readiness/program-*-preregistration.md` - the pre-registration for each program (e.g. `program-4-nondirectional-preregistration.md`, `program-8-weekly-crt-sweep-preregistration.md`).
- `CLAUDE.md` §6.5 - the Authority Ladder this whole chapter is an application of.

Unresolved questions

- **Program 9 run artifacts** - whether Stage-1/Stage-2 JSON under `results/research/m5_mtf/` is present and sealed into a finding was not re-validated against a fresh run this pass; the *design* is now chaptered, the *verdict* must still come from artifacts + findings.
- Program 7 (Open Interest) is named but its current status (planned, in progress, or not started) wasn't independently verified this pass.

Previous: [Chapter 18 - A Field Guide to docs/governance/](#) * **Next:** [Chapter 20 - The Research Platform](#)

Related: [Chapter 09 - Interpreters and the Pattern Contract](#) (the same discipline, one interpreter at a time) * [Chapter 16 - Config-First Doctrine](#) (the Authority Ladder)

Memory: none dedicated - see the research-family memory entries indexed in `MEMORY.md` under this session's project files (e.g. cross-sectional, carry-basis, carry-harvest programs).

Appendix A2 - Unresolved Questions (Rollup)

Status of this chapter: Written (updated Grok review pass, 2026-08-07)

Why this chapter exists

Per this book's own charter: "if understanding is impossible, do not invent." Every chapter that hit a genuine gap in evidence recorded it locally rather than guessing. This appendix collects those in one place - and, after the Grok review pass, also records what was **closed in the book** without changing production code.

What you need to already know

This is a reference appendix, not a narrative chapter - each item links back to the chapter that owns the full context.

Resolved in the book (Grok review pass - no code changes)

Item	Resolution	Chapter
INOUT <code>src/inout/</code> vs <code>archive/inout_legacy/</code>	Name collision. Current <code>src/inout/</code> = data fetchers only; archived tree = parallel execution rail retired 2026-05-02	Ch.15
Program 9 missing from research chapter	Chaptered from pre-registration (<code>docs/research/preregistration-program-9.md</code>); run verdicts still artifact-bound	Ch.19
ATR / F-057 "flag only" passivity	Proposed remediation options + trade-offs added; still OPEN pending owner	Ch.13
Measurement Contract gap	5-step seal path proposed; still 0 sealed instances in-repo unless registry says otherwise	Ch.20
MSIP "no treatment"	Orientation brief in governance field guide	Ch.18
Density / meta overwhelm	Quick Start chapter added	Ch.00
22 un-chaptered packages	Orientation map added (not full chapters)	Ch.04

Still open - book-scope gaps

- **Repository Encyclopedia** - charter in [Chapter 24](#);
E0-E6 + E1b + JSONL twin DONE (`docs/book/encyclopedia/`, including `encyclopedia_rows.jsonl`). Optional later: relationship graph edges in JSONL; agent query UI.
- **Full-depth Part VI-VIII** - chapters 16-23 remain orientation-level relative to their source corpora (~250 governance files, ~475 research files). Deepening is expected future book work.
- **Optional MSIP dedicated chapter** - brief exists; full gate-by-gate narrative only if MSIP re-enters active execution.
- **Program 9 numeric verdict** - design is chaptered; Stage-1/Stage-2 results must still be read from artifacts / findings, not assumed.

- **Program 7 (Open Interest)** - named as out-of-scope for Program 6; status not independently verified as run.
- **Deeper Part VIII implementation walk** - agent/control-plane/multi-LLM chapters improved slightly but are not full replacements for docs/reference/agent-reference.md.

Still open - real repository issues (code/doc; not fixed by this book pass)

User instruction for this pass: **no code changes**. These remain tracked, not remediated.

- docs/reference/schemas.md §4 still documents the 38-dim v3.0 feature tuple; live code is 39-dim v4.0. Textbook DOC_DRIFT. See [Chapter 7](#). *Recommended separate Doc Sync turn - not part of this PDF/book narrative pass.*
- docs/reference/testing.md file-count header ("116 files across 15 directories," dated 2026-06-12) is stale vs ~1,200+ files under tests/. See [Appendix A1](#).
- ATR unit mismatch in compute_crt_levels - real geometry defect; options A/B/C in [Chapter 13](#); awaiting owner decision.
- F-057 crtconfig programmatic split-brain - OPEN; options 1/2/3 in Chapter 13; "backtest Phase 2."
- Research Measurement Contract - schema frozen, 0 sealed MC-* (verify registry); path in [Chapter 20](#).

Verification gaps (assumed from docs; not re-proven this pass)

- Whether Wyckoff / Market Profile / Order Flow interpreters have code beyond scaffolding ([Chapter 9](#)).
- Whether every tool in docs/reference/agent-reference.md is end-to-end reachable ([Chapter 21](#)).
- Full body of docs/reference/architecture.md beyond header/orientation ([Chapter 4](#)).

Out of scope for this pass (explicit)

- Fixing schemas.md / testing.md primary docs (Doc Sync turn).
- Implementing ATR or F-057 remediations (code + parity + promotion).
- Sealing MC-* instances or running Program 9 Stage drivers.
- Adding CLAUDE.md companion-row (already present from an earlier session).

Previous: [Appendix A1 - Testing the System](#) * **Next:** - (last chapter; return to [the Table of Contents](#))

Related: every chapter linked above.

Memory: none dedicated - this appendix is the book's own open-loop tracker.